Behavior Aisle

Test a store change on simulated shoppers before moving a single box.

BehaviorGPT predicts each shopper's next action; paired simulation prices the change.

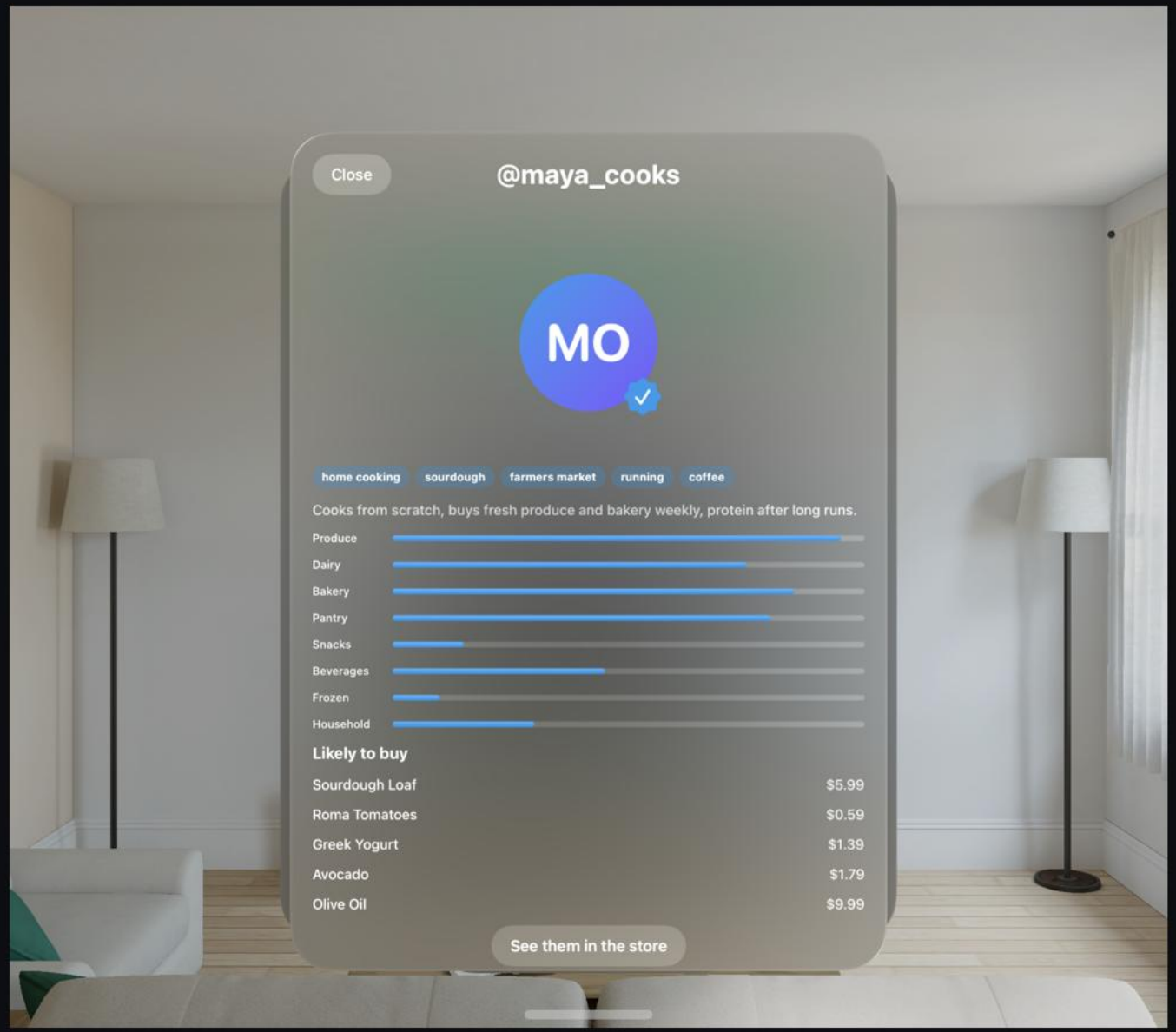
Apple Vision Pro, 1:1 grocery store.



1. Shoppers you can watch decide

Each HumanHD shopper walks a full-scale store; the panel shows its top predicted items and why.

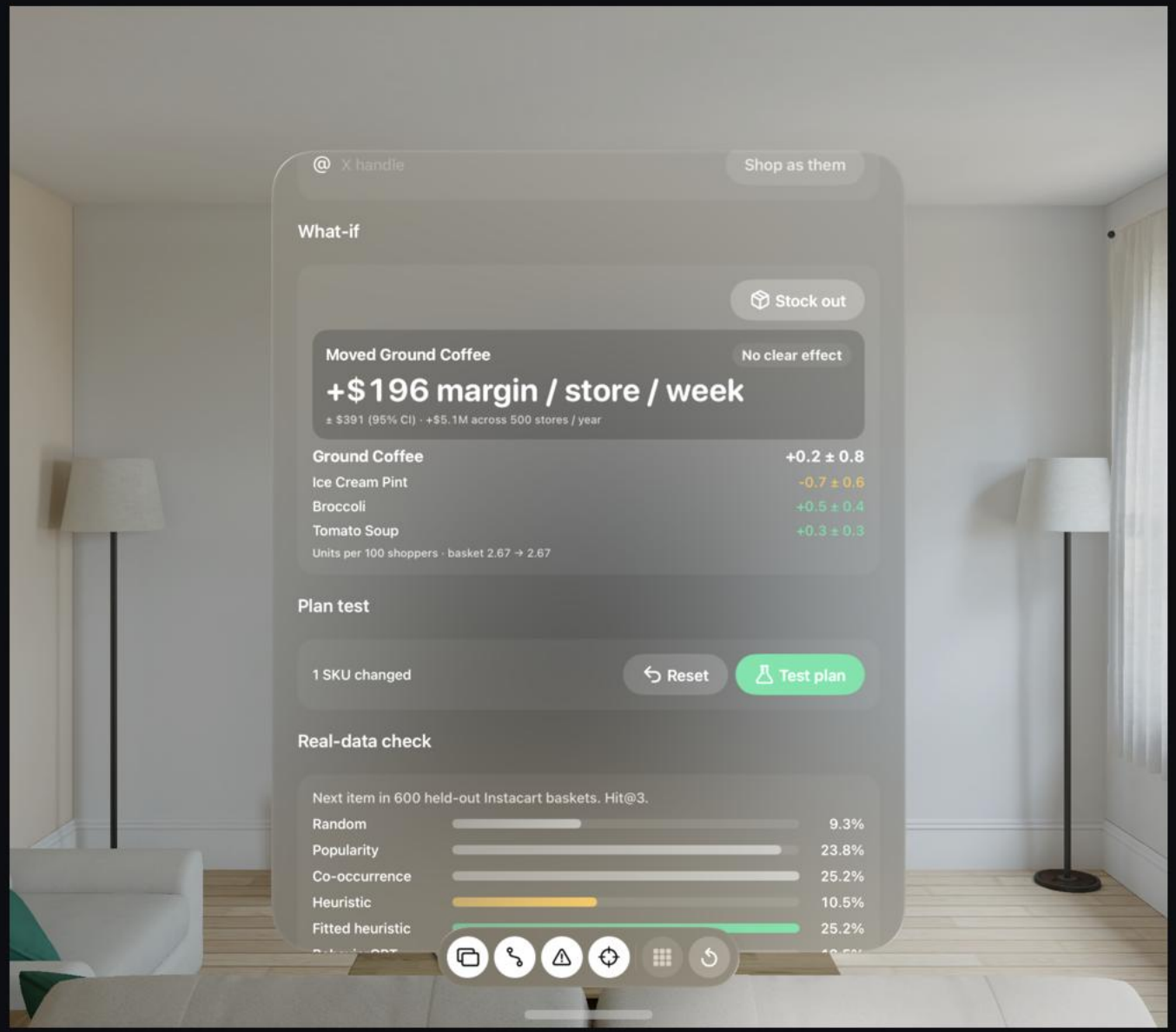
@maya_cooks: Bananas 10%, Red Apples 8%, Sourdough 7%, each traced to an interest.



2. A shopper from a real profile

Bio and posts become category affinity and a likely-to-buy list.

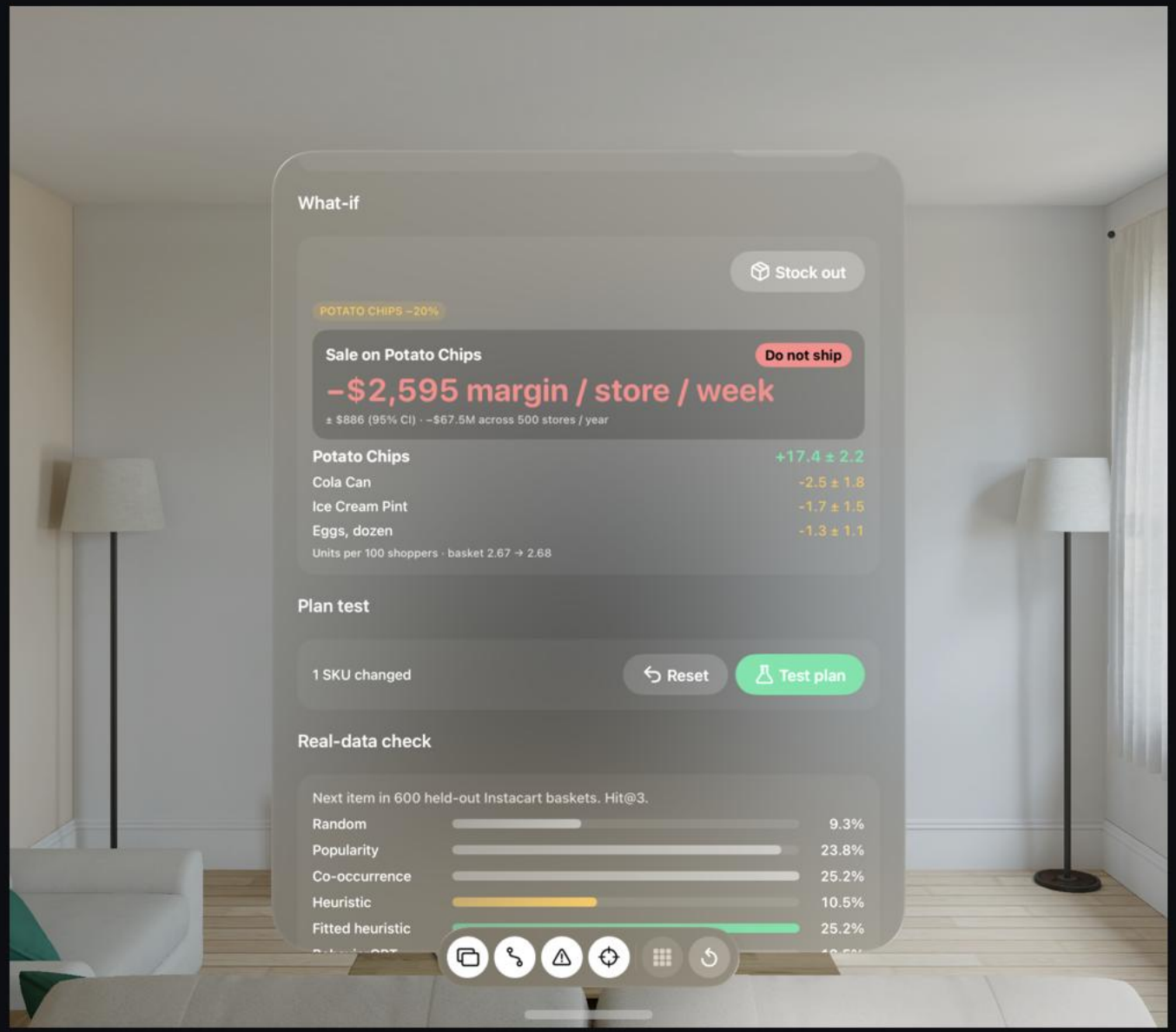
Sourdough, tomatoes, yogurt, avocado, olive oil for a home cook who runs.



3. Move a product, read the answer in dollars

Coffee moved to the front of aisle 3: +$196 margin / store / week, but ±$391.

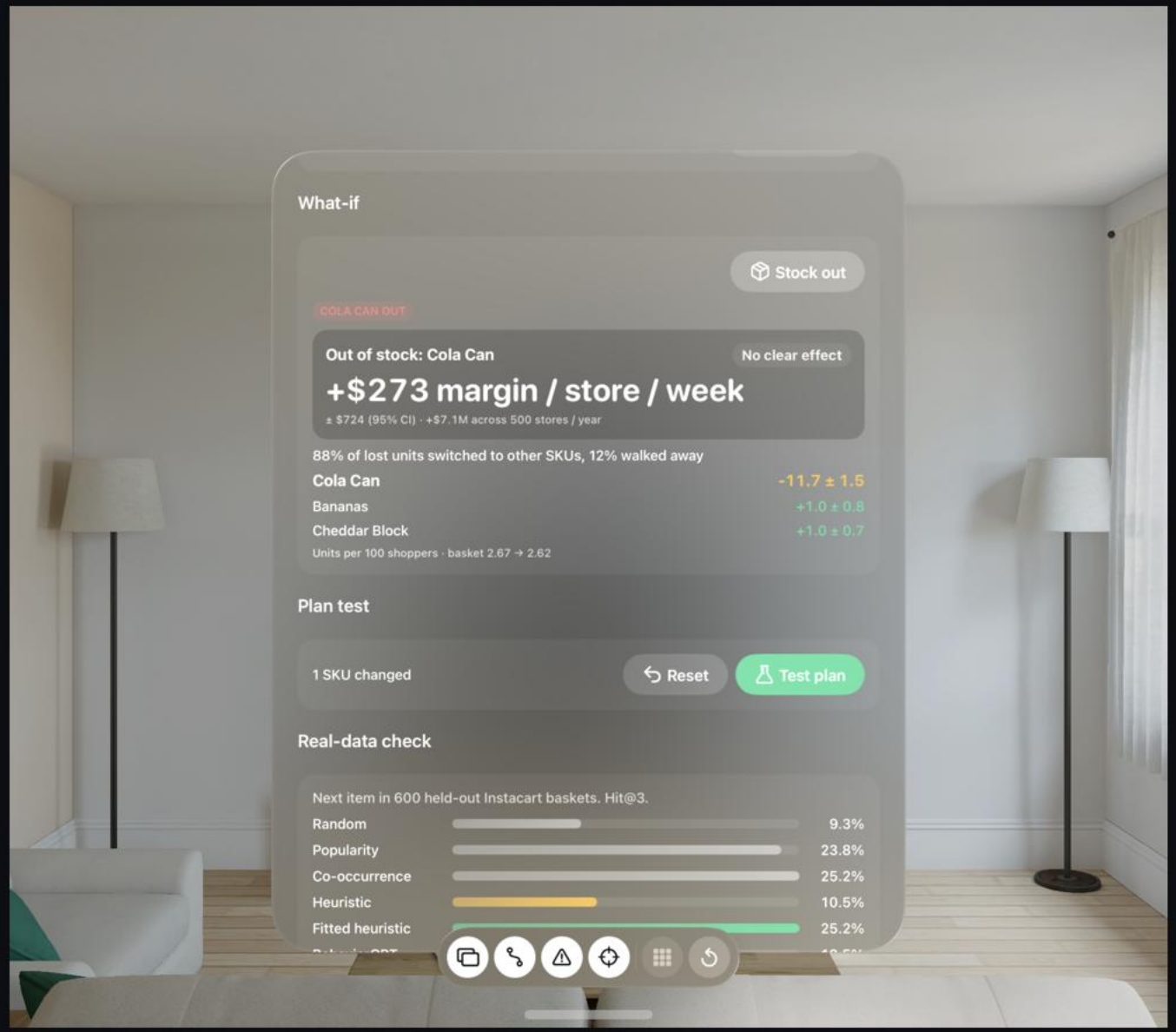
Verdict reads No clear effect. The tool refuses to call noise a win.



4. A promotion that loses money

Chips 20% off: +17.4 units per 100 shoppers, almost double.

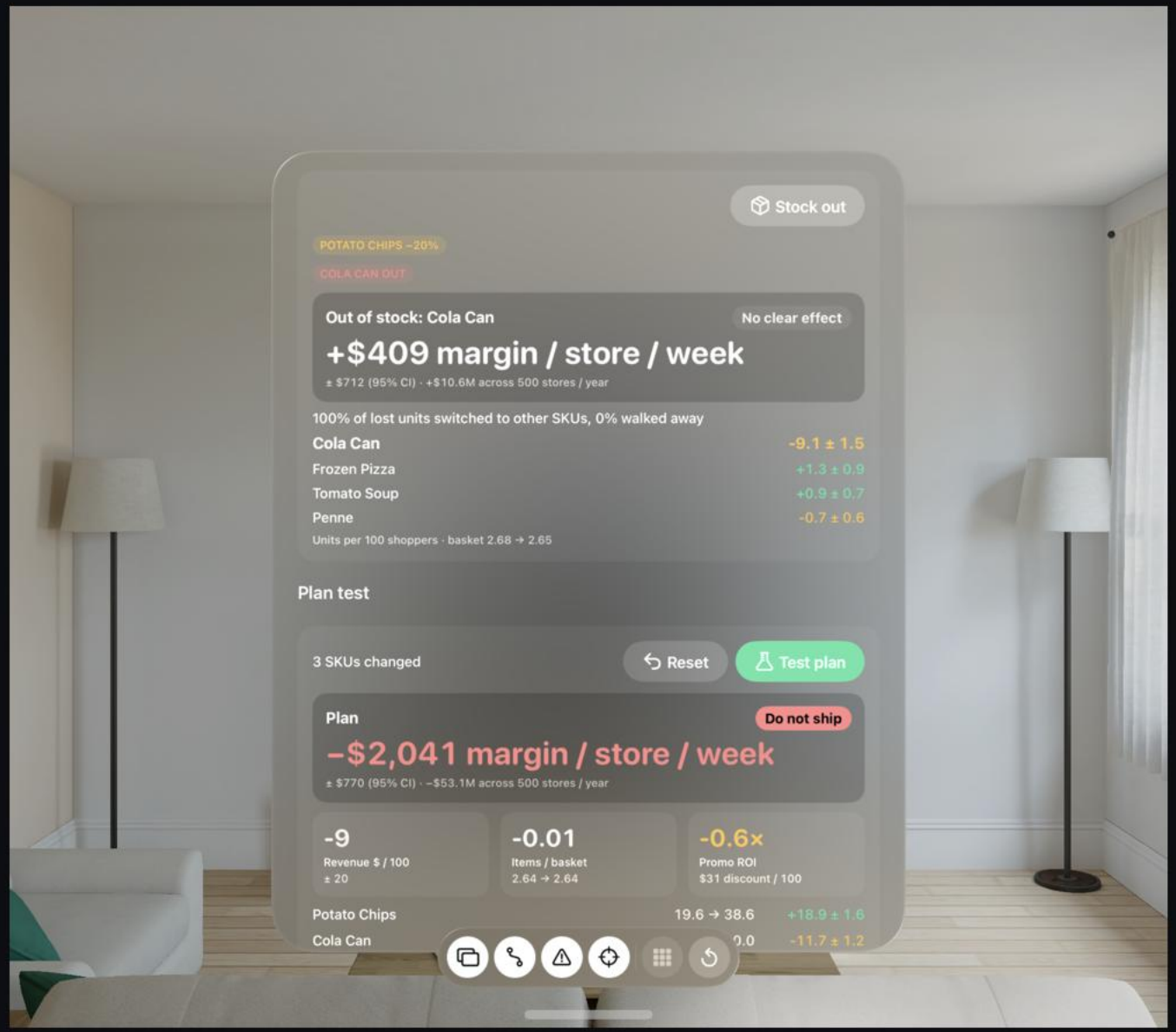
Margin: −$2,595 / store / week (±$886). Do not ship. Cola, ice cream lose units to chips.



5. Price a stock-out

Cola out of stock: 88% of lost units move to other SKUs, 12% walk away.

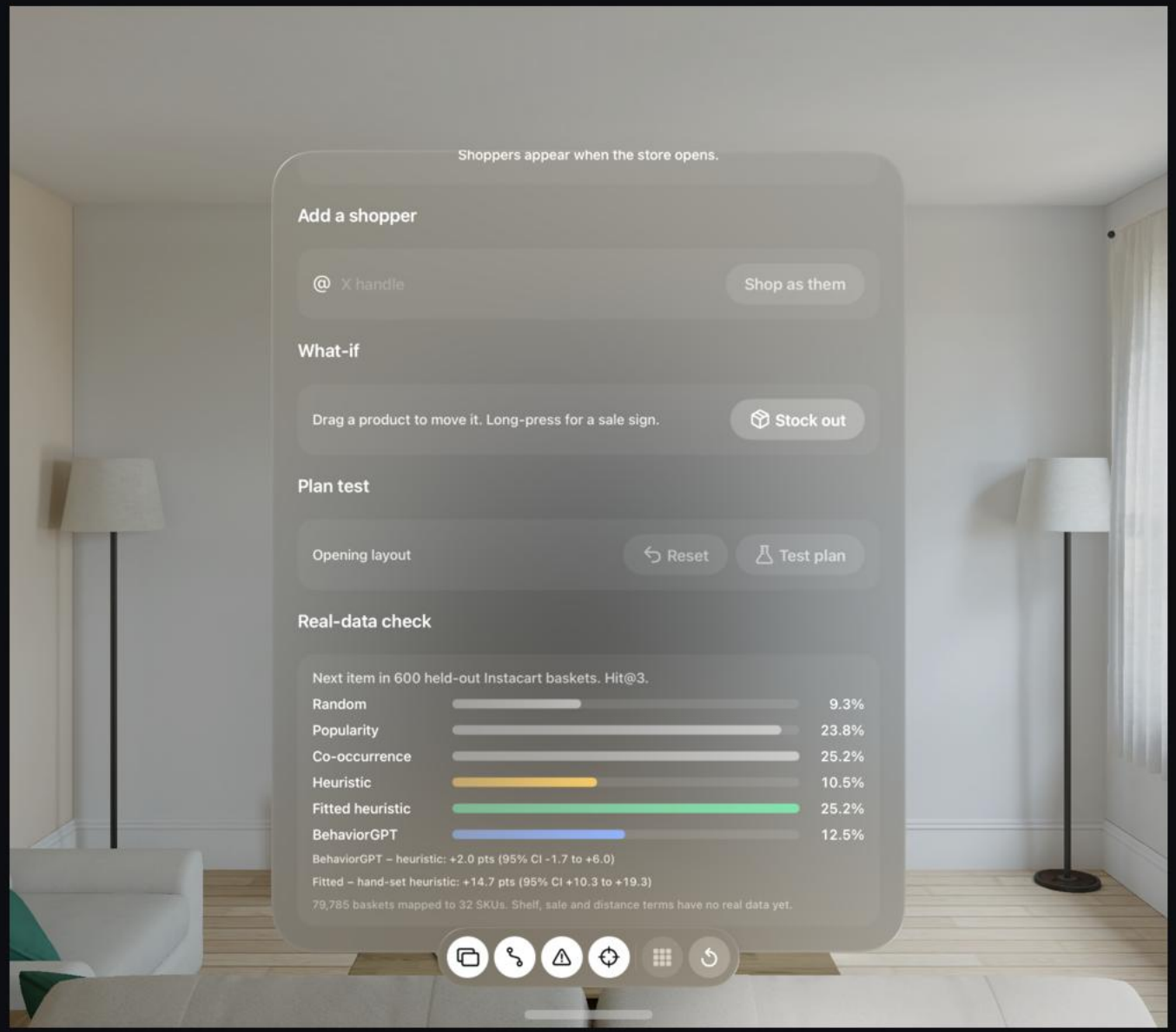
Tells a buyer which gaps cost money and which ones shoppers absorb.



6. Test the whole plan before reset night

Move + sale + stock-out together on 2,000 paired shoppers.

−$2,041 margin / store / week, −$53M a year across 500 stores. Promo ROI −0.6x.



7. Checked on real baskets

600 held-out Instacart baskets, next-item hit@3.

Fitted model 25.2% vs popularity 23.8% and random 9.3%. BehaviorGPT 12.5% on this mapping.

Assumptions in the numbers

Margins are category averages (produce 35%, dairy 25%, snacks 40% …); sale = 20% off, cost fixed.

12,000 shoppers / store / week, 500 stores. Effects from the fitted heuristic shopper model.

Shelf position, sale and walking-distance terms have no real-data fit yet.

BehaviorGPT drives live shoppers; paired simulation runs the heuristic for speed.